Acoustic Biot-Allard Parameters for Basotect Foam

Basotect is a melamine resin open-cell foam widely used as a sound-absorbing material. Its acoustic behavior is most comprehensively described by the **Biot-Allard model** (also called the Biot-Johnson-Champoux-Allard or BJCA model), which combines Biot's theory of elastic wave propagation in fluid-saturated porous media with the Johnson-Champoux-Allard (JCA) description of the fluid phase. This model requires up to **nine material parameters**: five transport (fluid-phase) parameters, two elastic (solid-frame) parameters, the bulk density, and additional thermal permeability (sixth parameter) if the full JCAL model is used.

Below is a detailed summary of the measured and predicted Biot-Allard parameters for Basotect (melamine) foam, synthesized from experimental studies.

1. Fluid-Phase (Transport) Parameters

The Johnson-Champoux-Allard (JCA) / JCAL model describes visco-thermal dissipation in the pore fluid. The five core parameters, plus the optional static thermal permeability, have been extensively characterized for Basotect.

1.1 Porosity (ϕ)

Basotect foam is highly porous. The porosity is typically estimated from the bulk density ρ_1 and the skeleton density ρ_m ($\rho_m = 1570$ kg/m^3 for melamine) via:

$$\phi = 1 - \frac{\rho_1}{\rho_m}$$

For uncompressed Basotect TG with bulk density ∼8.8–10.3 kg/m^3, porosity values range from **0.992 to 0.995** [1], [7]. For compressed samples, porosity decreases following [1]:

$$\phi^{(n)} = 1 - n(1 - \phi^{(1)})$$

where n is the compression ratio.

1.2 Static Airflow Resistivity (σ)

Flow resistivity is one of the most influential parameters for sound absorption [5]. For uncompressed Basotect TG (bulk density ∼8.8 kg/m^3), the flow resistivity is approximately **6200 Pa·s/m^2** [1]. For Basotect samples with higher bulk density, values increase to **10,500–17,500 Pa·s/m^2** [7]. For compressed samples, values can reach **122,325 Pa·s/m^2** at a compression ratio $n \approx 6.6$ [1], [2].

Kino et al. [1] established the following empirical relationship for Basotect TG:

$$\sigma_{\text{Kino(BTG)}} d^2_{\text{Kino(BTG)}} \rho_1^{-1.53} = 8.0 \times 10^{-9}$$

where d is the fiber-equivalent diameter (∼5.5–6.5 μm) and ρ_1 is the bulk density.

1.3 Tortuosity (α_∞)

Tortuosity describes the sinuosity of the pore paths. For uncompressed Basotect TG (sample 61), the measured tortuosity is **1.0101** [1]. For the Illtec melamine foam, values are similarly low (∼1.0053–1.0090) [7]. These low values indicate a very open, straight-pore structure. Upon compression, tortuosity increases gradually, reaching up to **1.073** at high compression rates [1], [2].

1.4 Viscous Characteristic Length (Λ)

The viscous characteristic length relates to the smaller pore dimensions where viscous dissipation dominates. For uncompressed Basotect TG (bulk density 8.77 kg/m^3), $\Lambda \approx$ **271 μm** [1]. For Illtec melamine foam samples (bulk density 10.3 kg/m^3), $\Lambda \approx$ **199 μm** [7]. Under compression, Λ decreases approximately as [1], [3]:

$$\Lambda^{(n)} = \frac{\Lambda^{(1)}}{n} + \frac{a}{2}\left(\frac{1}{n} - 1\right)$$

where a is the fiber-equivalent radius (~3 μm). At $n \approx 6.6$, Λ can drop to ~37–48 μm [1], [3].

1.5 Thermal Characteristic Length (Λ')

The thermal characteristic length is related to the total surface-to-volume ratio of the pores. For uncompressed Basotect, Λ' is approximately twice the viscous characteristic length: $\Lambda' \approx$ **572 μm** for Basotect TG [1]. For Illtec foam, $\Lambda' \approx$ **460 μm** [7]. The ratio Λ'/Λ generally satisfies $\Lambda'/\Lambda \geq 2$ [1], [6].

1.6 Static Thermal Permeability (k'_0) (JCAL model)

The full six-parameter JCAL model includes static thermal permeability. For uncompressed melamine foam, k'_0 is on the order of **4.0–6.3 $\times$ 10^{-9} m^2** [3]. Under compression, it decreases according to a relationship derived from fiber-array models [3].

2. Summary of Measured Values for Uncompressed Basotect TG

Parameter	**Symbol**	**Unit**	**Value (Ref. Sample 61)** [1]
Bulk density	ρ_1	kg/m^3	8.77
Surface density	-	g/m^2	173
Porosity	ϕ	-	0.9944
Flow resistivity	σ	Pa·s/m^2	6,197
Tortuosity	α_∞	-	1.0101
Viscous char. length	Λ	μm	271
Thermal char. length	Λ'	μm	572

For Illtec melamine foam (Sample 51, uncompressed) [1]:

Parameter	**Value**
ρ_1	12.39 kg/m^3
ϕ	0.9921
σ	10,943 Pa·s/m^2
α_∞	1.0090
Λ	230 μm
Λ'	460 μm

3. Elastic (Solid-Frame) Parameters

For poroelastic modeling under the full Biot model, the elastic properties of the melamine skeleton are required. Measurements show that Basotect foam is **orthotropic** (not simply isotropic) [4].

3.1 Young's Modulus (E)

The Young's modulus of melamine foam is strongly dependent on density and direction. For a typical uncompressed melamine foam of bulk density ∼9–10 kg/m^3, static Young's moduli in the compressional directions are on the order of **100–200 kPa** [6], [4]. Specifically, Cuenca et al. [6] determined the compressional moduli (elements of the Hooke's matrix) for melamine foam:

- $H_{11} \approx$ 1.2–1.3 MPa (highest stiffness direction)
- $H_{22} \approx$ 1.2–1.4 MPa
- $H_{33} \approx$ 3.7–3.8 MPa (lowest stiffness in the rise direction)

From a dynamic characterization, the Young's modulus for melamine foam used in impedance tube inverse methods is approximately **190–200 kPa** [6].

3.2 Poisson's Ratio (ν)

Poisson's ratio for open-cell melamine foams is typically in the range **0.3–0.44** [6]. For the purposes of one-dimensional impedance tube modeling, it is often set to zero because the lateral deformation cannot be independently resolved from a single normal-incidence measurement [6].

3.3 Loss Factor (η)

The structural damping of melamine foam is typically $\boldsymbol{\eta}$ = **0.04–0.12** [6]. The specific value depends on frequency and temperature.

3.4 Bulk Density (ρ_1)

The bulk density of Basotect foam is typically **8–12 kg/m^3** for uncompressed commercial grades [1], [7]. Under thermo-compression, it scales linearly with compression ratio: $\rho_1^{(n)} = n \cdot \rho_1^{(1)}$ [3].

4. Sensitivity and Model Selection

4.1 Which Parameters Matter Most?

A global sensitivity analysis by Ouisse et al. [5] using the Champoux-Allard (five-parameter) and Biot-Allard (nine-parameter) models revealed a strong frequency-dependent hierarchy:

- **Flow resistivity ($\boldsymbol{\sigma}$)** and **viscous characteristic length ($\boldsymbol{\Lambda}$)** are the most influential parameters across the audio frequency range for sound absorption.
- **Porosity ($\boldsymbol{\phi}$)** has limited impact on the absorption coefficient except at very low frequencies.
- **Tortuosity ($\boldsymbol{\alpha_\infty}$)** plays a secondary but non-negligible role, particularly around mid-frequencies.
- **Elastic parameters (E, $\boldsymbol{\nu}$, $\boldsymbol{\eta}$)** become critically important in frequency regions where the frame resonance (fluid-structure coupling) occurs. For standard 25–30 mm thick melamine foam, this is typically around **1350–1500 Hz** [6].

4.2 Rigid-Frame vs. Elastic-Frame Models

For many applications, Basotect can be modeled with the **rigid-frame Johnson-Champoux-Allard model** (5 parameters), as the frame motion is negligible at low frequencies or when the material is thick enough [1], [7]. However, when the foam is used in thin layers or when structural coupling is important (e.g., in sandwich panels), the full **Biot-Allard model** (9 parameters including E, ν, η, ρ_1) is required to capture the additional damping caused by the frame resonance [6], [5].

4.3 Compression Effects

Basotect foams are often thermo-compressed in automotive applications. The compression model proposed by Lei et al. [3] provides a comprehensive framework to predict all six JCAL parameters at any compression rate n from the initial (uncompressed) values. A key finding is that the **fiber/ligament orientation changes under compression**, significantly affecting σ and Λ but having weaker effects on α_∞ [3].

4.4 Cross-sectional Pore Shape Factors

The Johnson-Allard model introduces cross-sectional shape factors c and c'. For Basotect TG, these are approximately **c ≈ 0.25–0.57** and **c' ≈ 0.24–0.40**, which are significantly lower than those for glass wool (c ≈ 0.71–0.78) [7]. This explains how melamine foam achieves high flow resistivity despite having large characteristic lengths [7].

5. Typical Complete Parameter Set for Biot-Allard Modeling

Based on the reviewed literature, a representative complete set of Biot-Allard parameters for **uncompressed Basotect TG melamine foam** (bulk density ≈ 8.8 kg/m^3) is compiled below:

Parameter	**Symbol**	**Unit**	**Typical Value**	**Source**
Porosity	ϕ	-	0.994	[1]
Flow resistivity	σ	Pa·s/m^2	6,200–13,100	[1], [7]
Tortuosity	α_∞	-	1.005–1.010	[1], [7]
Viscous char. length	Λ	μm	200–270	[1], [7]
Thermal char. length	Λ'	μm	450–570	[1], [7]
Static thermal perm.	k'_0	$\times 10^{-9}$ m^2	4.0–6.3	[3]
Bulk density	ρ_1	kg/m^3	8–12	[1]
Young's modulus	E	kPa	190–200	[6]
Poisson's ratio	ν	-	0 (or 0.3–0.44)	[6]
Loss factor	η	-	0.04–0.05	[6]

6. Recent Advances in Parameter Estimation

Cuenca et al. [6] developed a powerful multi-observation inverse method that simultaneously estimates all nine Biot-Allard parameters (plus thickness) from two impedance tube measurements (rigid-backed and coupled to an expansion chamber). Applied to a melamine foam sample, this method yielded:

$h = 23.14$ mm, $\rho_1 = 11.87$ kg/m^3

$\phi = 1.00$ (near unity), $\sigma = 18.6$ kN·s/m^4, $\alpha_\infty = 1.001$

$\Lambda = 69.5\ \mu$m, $\Lambda' = 158.7\ \mu$m

$E = 194$ kPa, $\eta = 4.6\%$

These values demonstrate the consistency of the Biot-Allard framework when properly applied to melamine foam characterization.

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